Practical Work: Text Clustering and Evaluation Report

Taha Mahha

March 17, 2026

1 Introduction

This report summarizes the implementation and evaluation of an unsupervised K-Means clustering pipeline applied to textual data. The objective was to categorize documents using TF-IDF vectorization, determine the optimal hyperparameter k, and validate the resulting clusters against ground-truth labels.

2 Dimensionality Reduction and Clustering

To visualize the high-dimensional TF-IDF sparse matrix, the data was projected onto two principal components using Truncated SVD. The K-Means algorithm was iteratively applied to group the texts. Figure 1 illustrates the spatial distribution and cluster assignments for $k \in \{2, 3, 4, 5\}$.

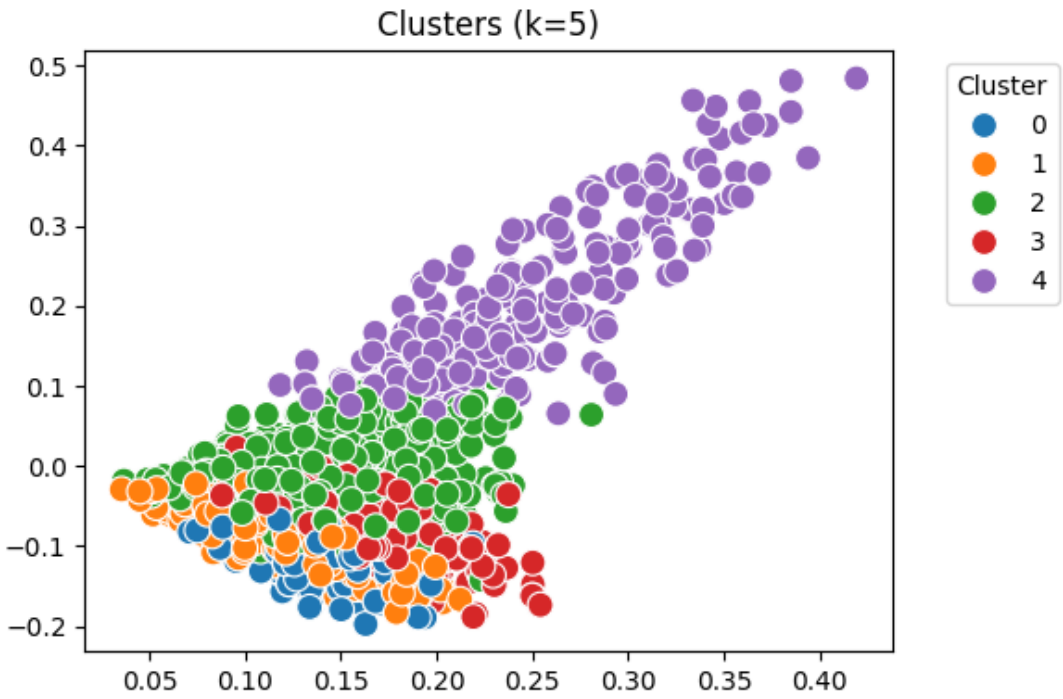

Figure 1: 2D Visualization of K-Means clusters ($k = 2$ to $k = 5$) using Truncated SVD.

3 Hyperparameter Optimization

To objectively determine the optimal number of clusters, the within-cluster sum of squares (Inertia) and the Silhouette Score were computed across multiple iterations.

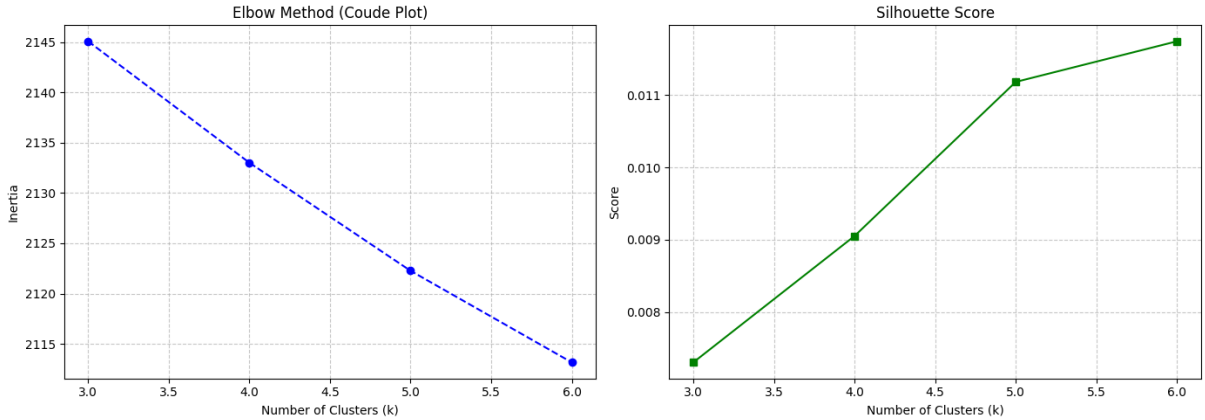

Figure 2: Elbow Method (left) and Silhouette Score (right) evaluating cluster configurations.

As shown in Figure 2, while the inertia plot exhibits a smooth linear descent without a sharp "elbow," the Silhouette Score provides a definitive, absolute maximum at $k = 5$. This peak mathematically confirms that 5 clusters yield the strongest balance of intra-cluster cohesion and inter-cluster separation.

4 External Validation

Having established $k = 5$ as the optimal unsupervised configuration, the resulting clusters were cross-referenced with the true binary ground labels (Category 0 and Category 1) to assess real-world alignment.

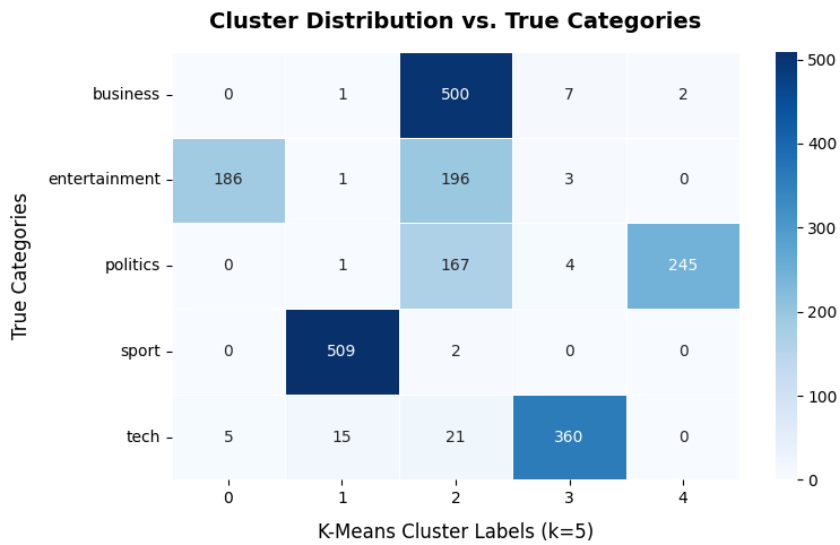

Figure 3: Heatmap of K-Means Cluster Labels ($k = 5$) compared to True Categories.

Figure 3 demonstrates robust predictive capabilities within the model:

- **Cluster 4** is highly pure, isolating Category 1 (705 documents) with very minimal overlap from Category 0 (40 documents).

- **Cluster 3** effectively captures the dominant theme of Category 0, containing 874 instances.
- **Clusters 0, 1, and 2** contain a more balanced mix of both categories. In text mining, this indicates shared vocabulary, neutral statements, or overlapping semantic sub-topics that bridge the two primary sentiments.

5 Conclusion

The application of K-Means clustering at $k = 5$ successfully uncovered latent semantic structures within the unstructured text. The external validation confirms that the mathematically optimal clusters closely map to the true underlying categories, isolating the core sentiments into highly pure groups while separating nuanced, mixed-topic documents into distinct clusters.